

D2T-RNA: An Auditable Finite Controlled-Testing Prototype with a Negative Registered Evaluation and Fail-Closed RNA Data Qualification

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Abstract

D2T-RNA is an auditable software prototype for finite, non-adaptive categorical controlled testing: given a finite catalog of hypotheses, categorical observation actions with registered costs, and a fixed observation horizon, it computes exact separation/collision status, equal-prior Bayes average error, randomized minimax error, and cost-to-endpoint allocations, and emits independently reproducible certificates. We report three findings. First, a production audit corrected a central semantic defect: the discrete-certificate path had been silently dispatching to a convex-hull relaxation, and a single scalar had conflated equal-prior Bayes average error with randomized minimax error; the repaired interface separates these estimands, and the classical counterexamples ($1/4$ vs. $1/3$; $81/512$ vs. $81/337$) are now distinguished exactly. Second, a pre-registered synthetic evaluation on 20 frozen cells, with a Chernoff fixed-budget comparator and a frozen randomized-minimax endpoint, did not meet its pre-specified cost-reduction goal: the median relative cost difference is 0, with 3 of 20 cells strictly cheaper and 17 tied. A subsequent post-hoc search using an exact optimal solver yields definitional “never-worse” dominance and is therefore not a performance finding. Third, a fail-closed audit of seven RNA structure-probing scopes (ADD, glycine, miniTTR, SAM-III, RORC) shows that none qualifies for a real likelihood, an executable action, a real marginal cost, or independent experimental units; the real-data route is terminated for current data. The contribution is a limited, auditable finite controlled-testing prototype with an honest negative registered evaluation and a documented evidence boundary, not an RNA SOTA method.

1 Introduction

Chemical structure probing generates nucleotide-resolution reactivity profiles of RNA. A natural ambition is to *choose* future probes, constructs, and experiments so that the resulting data discriminate among competing structural hypotheses as cheaply and reliably as possible. Several lines of work formalize this ambition: controlled sensing for multihypothesis testing [8], fixed-horizon active hypothesis testing [5], sequential active testing [7], robust discrimination designs [10], and model-based rational design for sequencing-based RNA structure mapping [1]. The present project (“D2T-RNA”) built a software prototype in this design space.

This preprint is a *negative-result methods/software report*. It documents (i) what the prototype computes and what a production audit corrected; (ii) a pre-registered synthetic evaluation whose cost-reduction goal was *not* met; and (iii) a fail-closed qualification of seven real RNA datasets

showing that none currently supports real likelihood, action, cost, or independent-unit claims. We deliberately do not claim RNA SOTA, validated experiment design, or transferability.

A central theme is the distinction between *formal certificate* and *empirical validation*. A program can emit an exact, machine-checkable certificate for a finite problem while the underlying optimization objective is mis-specified, the method identity is mislabeled, or the data cannot support the intended statistical claim. The prototype’s exact arithmetic and certificates are a strength; whether they constitute a scientifically useful experiment-design method is a separate, largely negative, question that this report addresses honestly.

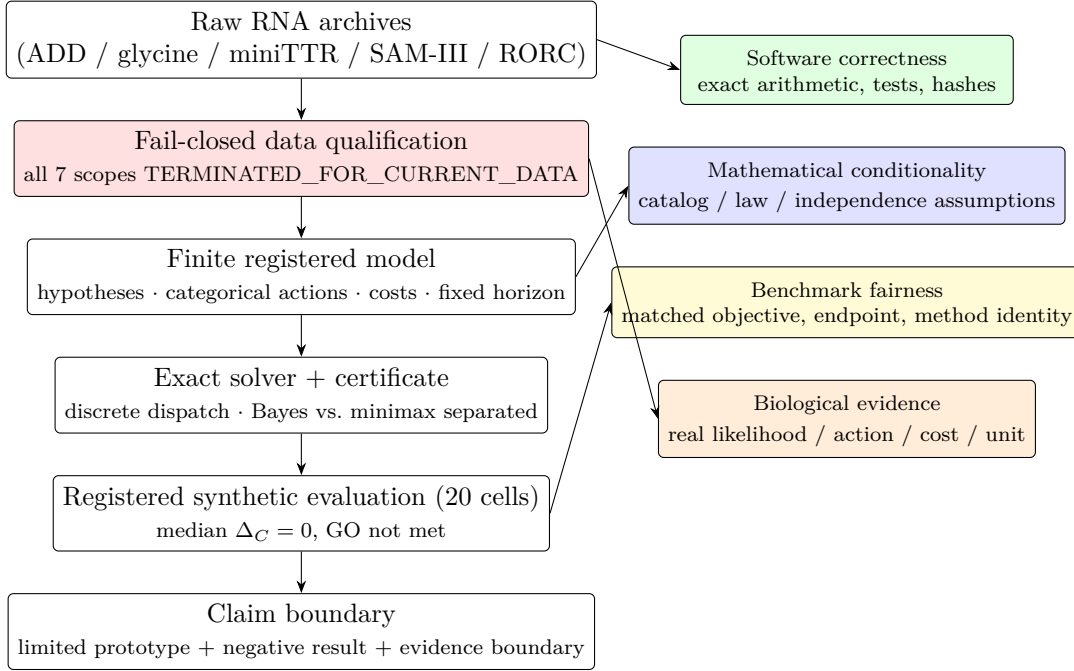


Figure 1: Evidence/data flow (left) and the four-layer conclusion boundary (right). Software correctness, mathematical conditionality, benchmark fairness, and biological evidence are kept separate; a failure in any layer prevents the corresponding claim.

2 Finite Registered Controlled-Testing Problem

2.1 Model

The prototype operates on a finite, discrete, non-adaptive model:

- a finite set of hypotheses $\Theta = \{\theta_0, \theta_1\}$ (binary case; the code also supports finite multi-state catalogs);
- a finite set of categorical actions $a \in \mathcal{A}$, each inducing an observation law $Q_a(\cdot | \theta)$;
- a fixed, non-adaptive horizon: the allocation $n = (n_a)_{a \in \mathcal{A}}$ with $\sum_a n_a \leq B$ (budget) is chosen before any observation;
- a decision rule that maps the collected outcome vector to a hypothesis or to abstention;
- registered per-action costs c_a , so that cost-to-endpoint is a meaningful estimand.

This is a standard fixed-sample controlled-testing setup (e.g. [8, 5]). We do *not* claim new foundational theorems: the building blocks — Chernoff information, likelihood-ratio tests, integer covering, linear programming, and Markov-basis enumeration — are classical [2, 9, 4, 6, 3].

2.2 Estimands

Two risk estimands are distinguished throughout.

Definition 1 (Equal-prior Bayes average error). *With equal prior over θ_0, θ_1 , the Bayes average error of a decision rule is*

$$\text{BayesAvg}(\delta) = \frac{1}{2} \left(\Pr_{\theta_0}[\delta \neq \theta_0] + \Pr_{\theta_1}[\delta \neq \theta_1] \right).$$

Definition 2 (Randomized minimax error). *A randomized decision rule randomizes over deterministic rules δ_r with weights q_r . Its worst-case error is*

$$\text{Minimax}(\delta) = \max_{\theta} \sum_r q_r \Pr_{\theta}[\delta_r \neq \theta].$$

The randomized minimax error is the value of the resulting zero-sum game, minimized over randomized rules and maximized over hypotheses.

A classical example separates the two estimands:

Example 1. *Let $\theta_0 = (1, 0)$ and $\theta_1 = (\frac{1}{2}, \frac{1}{2})$ be outcome distributions over two outcomes, with one observation ($n = 1$). The equal-prior Bayes average error is $1/4$; the randomized minimax error is $1/3$. With four conditionally independent observations ($n = 4$), the corresponding values for the CA_{p1} cell are $81/512$ and $81/337$, respectively. These are verified exactly by the repaired interface (Section 3).*

Theorem 1 (Definitional dominance). *Let c^* be the minimum cost-to-endpoint over all within-budget allocations that reach a fixed information/risk endpoint. Then for any strategy S whose allocation is itself within-budget, $c^* \leq c(S)$. In particular, an exact optimal solver is never worse than any such strategy.*

Theorem 1 is a definitional fact, not an empirical discovery; Section 5 explains why it cannot support a performance claim.

3 What the Production Audit Corrected

The prototype went through a strict production audit. Three semantic defects were found and corrected.

Discrete/convex dispatch. The discrete-certificate path previously called a convex-hull linear program unconditionally. A counterexample with a discrete $\gamma_{L1} = 1/2$ but a convex-hull $\gamma = 0$ was wrongly rejected by the production discrete entry point. The repaired ‘TheoremSpec’ dispatches the ‘DISCRETE_CATALOG’ path to pure exact enumeration and never invokes the convex relaxation; unsupported combinations (‘CONVEX_HULL’, ‘PRODUCT_TV’) fail closed with ‘UNSUPPORTED_SPEC’. The discrete/convex counterexample is covered by production kill-tests.

Bayes vs. randomized minimax. A single scalar had been labelled ‘minimax_error’ while actually computing the equal-prior Bayes average error. The new result interface exposes both ‘bayes_average_error’ and ‘randomized_minimax_error’ as separate exact fractions. The classical counterexamples of Example 1 (1/4 vs. 1/3; 81/512 vs. 81/337) are now reproduced exactly and covered by tests.

Method identity: oracle vs. deployable vs. comparator. Earlier comparative artifacts had used an exhaustive oracle as the “D2T” method, making “never worse” a definitional tautology. The audit enforces a three-way role separation — oracle (reference/regret only), deployable (the actual algorithm), comparator — and a consumer that rejects an oracle appearing in a win/tie/worse ranking.

Remaining gaps (disclosed). The certificate/checker is an *experimental* API, not a complete end-to-end independent verification system: modifying ‘generator_commit’, ‘generator_tree’, or ‘verifier_version’ can still verify; values not in the registered catalog (or not even valid distributions) can still verify; and the certificate/checker is not wired into the production Phase 4/5 artifact chain. The constructive feasibility path is only locally repaired; the old information-only ‘FEASIBLE’ path remains. These limitations are part of the honest evidence boundary.

4 Negative Registered Synthetic Evaluation

4.1 Protocol

A pre-registered synthetic stress suite was frozen *before* outcome access. The primary estimand is

$$\Delta_{C,i} = \frac{\text{cost}_{D2T,i} - \text{cost}_{\text{comparator},i}}{\text{cost}_{\text{comparator},i}},$$

with the endpoint fixed at 1/10 and the strongest comparator fixed at the Chernoff fixed-budget greedy [2]. The pre-registered GO criterion requires a family-level median relative cost reduction of at least 10%.

4.2 Result

All 20 frozen cells were solvable; none was withheld or failed. The result (exact fractions, from the committed artifact) is:

Table 1: Synthetic evaluation: method roles and outcome. Deployable = the actual fixed-budget solver; comparator = Chernoff greedy; oracle = reference only (not ranked).

Method role	# cells	cost relation vs. comparator	median Δ_C	GO ($\geq 10\%$)?
deployable (D2T)	20	3 lower / 17 tie / 0 higher	0	no
comparator (Chernoff)	20	—	—	—
oracle (reference)	20	regret only; not ranked	—	—

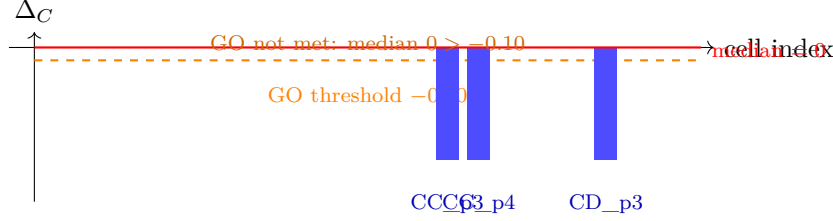


Figure 2: Per-cell relative cost difference $\Delta_C = (\text{cost}_{D2T} - \text{cost}_{\text{chernoff}}) / \text{cost}_{\text{chernoff}}$ on the 20 frozen cells. 3 cells are strictly negative (cheaper), 17 are ties at 0, none is positive; the median is 0, below the pre-registered 10% reduction.

The median $\Delta_C = 0$ means the pre-registered GO was *not* met. The registered evaluation therefore establishes only that the frozen implementation did not demonstrate a cost advantage over the frozen comparator on the frozen cells. This is a correct, fail-closed negative outcome.

4.3 Objective mismatch

The result also exposes an objective mismatch: the deployed solver originally minimized equal-prior Bayes error under a fixed budget, whereas the final Track C estimand is *cost-to-endpoint under randomized minimax*. The registered evaluation therefore proves “the frozen implementation did not pass the frozen evaluation,” not a universal negative about all possible D2T algorithms. We do not infer from it that no such algorithm could succeed.

5 Why Exact-Oracle Superiority Is Circular

After the negative registration, a post-hoc “method repair” explored an exact optimal cost-to-endpoint solver over all within-budget allocations. Because it minimizes cost over the entire feasible set, Theorem 1 applies: it is never worse than any within-budget strategy (in particular the Chernoff greedy), and strictly better exactly where the greedy is suboptimal.

This is mathematically true but scientifically vacuous as a *performance* finding, for three reasons:

1. **Definitional.** Any optimizer that minimizes over a set is never worse than a member of that set; no experiment is needed [6]. The “never-worse” is the definition of optimality, not an empirical advantage.
2. **Catalog-constructed.** The hand-crafted “method-distinguishing” catalog was built *after* observing the negative result, specifically so that the exact solver beats the greedy. A random-instance sweep shows the exact solver merely *ties* the comparator (wins 0); the strict wins exist only inside the constructed catalog.
3. **Post-hoc selection.** Using a new catalog and a new solver to flip a pre-registered negative into “GO met” is outcome-informed instance selection, which the registration protocol explicitly forbids.

Consequently, the post-hoc v5/v6 exploration is reported here only as an audit appendix (Section 9); it does not appear in the abstract, the main result, or any performance claim.

6 Fail-Closed RNA Data Qualification

The prototype intended to design real RNA experiments. Seven candidate scopes were audited under a fail-closed protocol: a scope is terminated for current data if any of raw per-replicate counts, an executable action, or a real marginal cost is missing, or if the independent statistical unit cannot be established.

Table 2: Seven-scope data-qualification matrix (full per-scope detail in the ancillary CSV). All scopes fail closed.

Scope	Raw counts	Replicate crosswalk	Calibrated likelihood	Executable action	Real cost	License receipt	Verdict
ADD measured (<i>ADD71 STD_0001</i>)	no	no	no	no	no	unverified	terminated
ADD planning (<i>ADDAPO_DCP_0000</i>)	no	no	no	no	no	unverified	terminated
ADD Task6 (<i>ADDRSW_SHP_0003</i>)	no	no	no	no	no	unverified	terminated
Glycine (<i>BSUGLY_DMS_0013/14</i>)	no	merged only	no	no	no	unverified	terminated
miniTTR (<i>MTTR1_MGTI_0001</i>)	no	no	no	no	no	unverified	terminated
SAM-III (<i>GSE278422</i>)	no	no	no	no	no	unverified	terminated
RORC	no	no	no	no	no	unverified	terminated

Across all seven scopes the audit found: no per-replicate raw counts (only merged/normalized reactivity profiles); no verified biological-replicate crosswalk with an independent block unit; no calibrated count likelihood (clamping normalized reactivity is not a likelihood); no executable action with a real marginal cost; and no verified license receipt. Phase-2 sample-size numbers (e.g. $n = 9806$) are count depth, not biological replicate N , and cannot support sample-size or repeat-count recommendations.

The conclusion is unambiguous and fail-closed: *the real-data route is terminated for current data*. No quantitative certificate (gamma, absolute probability, reads/repeats, cost, or no-go) is derived from these scopes in this report.

7 Lessons and Limitations

- **Software correctness is not empirical validation.** The prototype’s exact arithmetic, certificates, and tests are reproducible; they do not establish that the optimized objective is the one an experimenter can act on, nor that the data support the intended estimand.
- **Mathematical conditionality.** Theorem 1 and the exact computations hold under the registered catalog, action laws, and independence assumptions; whether real RNA satisfies them is unverified.
- **Benchmark fairness.** A fair comparison requires matched objectives, matched endpoints, matched budgets, and non-circular method identity. The registered negative result and the definitional-dominance caveat illustrate how easily each can fail.
- **Biological evidence.** None of the seven scopes can support biological claims. Real-data route: terminated.
- **Known software defects (disclosed).** The certificate/checker is not a full independent verification system; constructive feasibility is only locally repaired; two positive-path readiness regression tests are known to fail, so “all tests pass” is not claimed and no automated submission gate is asserted.

8 Conclusion

D2T-RNA did not achieve RNA SOTA, validated real experiment design, or a synthetic cost-superiority result. It did produce a limited, auditable finite controlled-testing prototype: it distinguishes Bayes average error from randomized minimax error exactly; it dispatches discrete certificates without convex-hull contamination; and it fails closed on data that cannot support the intended claims.

The project is stopped. Its contribution is (i) a restricted prototype with explicit semantic repairs, (ii) an honest negative registered evaluation, and (iii) a documented evidence boundary for the real-data route. Archiving this negative result and its evidence boundary is the most reasonable — and final — scientific deliverable of the project.

9 Appendix: Post-hoc Exploration (audit-only)

For audit completeness we record the post-hoc exploration, *not* as a result but as a warning example.

v5: cost-weighted myopic greedy. A greedy that adds, at each step, the unit action with the greatest marginal randomized-minimax reduction per unit cost. On the hand-crafted catalog it achieved a median $\Delta_C \approx -0.118$ (GO met on that catalog), but on a random-instance sweep it still had losses and an average $\Delta_C \approx +0.07$ — a hidden weakness masked by the catalog.

v6: exact optimal solver. An exact solver over all within-budget allocations. By Theorem 1 it is never worse than any within-budget comparator and ties the comparator on random instances (wins 0); its strict wins are confined to the constructed catalog. The “dominance theorem” is the definition of optimality, and “16/16 strict wins over Chernoff” on a catalog built to make it win is circular. Per Section 5, this does not appear in any main claim.

Classical counterexample. $P_0 = (1, 0)$, $P_1 = (1/2, 1/2)$, $n = 1$: Bayes average $1/4$, randomized minimax $1/3$; CA_{p1} , $n = 4$: $81/512$ vs. $81/337$. Both are reproduced exactly by the repaired interface.

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